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Strategic use of AI in Research

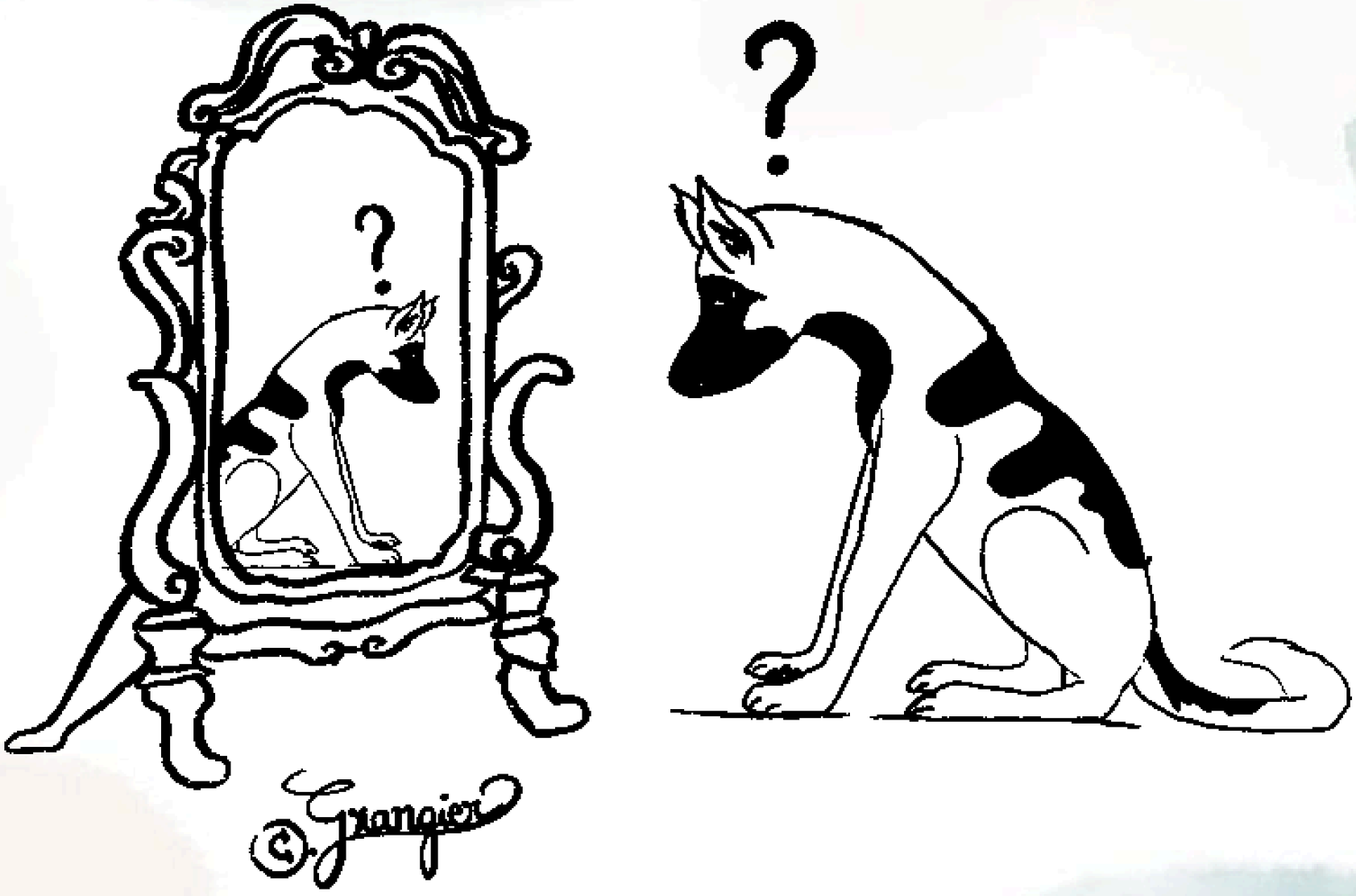
Kumar Aniket

Associate Professor in Infrastructure, Automation and Economic Transformation
Institute for Connected Communities
University of East London

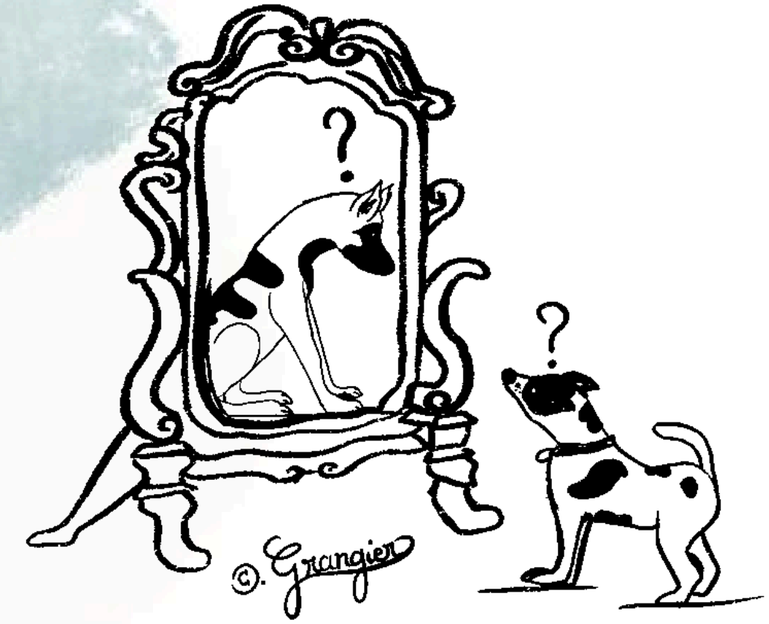
Director of Studies for Economics
Wolfson College
University of Cambridge

11 November 2026

Perception about AI



What is good Research?



How has AI changed research?

- What is the space within which research teams compete with each to produce research

AI + Human

Question



How do you cross the road as a pedestrian?

What is your research methodology?

Question

What do you we already know?

- Implicit knowledge
  - Polyani
- Explicit knowledge
  - Encapsulated in the academic literature

What is the research hypothesis?

- What do we not know collectively as a human?
  - Don’t try to be a journalist

Research Methodology

- Strength and weakness of competing research methodologies? Single domain expertise is a big risk factor with AI



- *How do you cross the road as a pedestrian?*
  - What is your research methodology?

Good research

Good research is **not** about convincing **anyone**

Good research is about **convincing yourself** that you have learnt something that you genuinely did not know before you started the research

Moykr’s Knowledge

Mokyr

- “**Useful knowledge**”
  - knowledge that helps people **understand** or **manipulate** the world around them.

Moykr: Prescriptive Knowledge

Prescriptive Knowledge (<math display="inline" xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mi>λ</mi><annotation encoding="application/x-tex">\lambda</annotation></semantics></math>-knowledge):

- Knowledge of **how** to do something.

It is the **recipe**, **technique**, or **invention** itself.

- Example:
  - Adding a spoonful of starter curd to warm milk will transform it overnight. No understanding of microbiology is required; one simply has to follow the steps.

Moykr: Prescriptive versus Propositional Knowledge

Prescriptive Knowledge (<math display="inline" xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mi>λ</mi><annotation encoding="application/x-tex">\lambda</annotation></semantics></math>-knowledge):

- Knowledge of **how** to do something.

It is the **recipe**, **technique**, or **invention** itself.

- Example: Yoghurt
  - Adding a spoonful of starter yoghurt to warm milk will transform it overnight. No understanding of microbiology is required; one simply has to follow the steps.

- Example: Social Networks
  - How do social networks work over long distance?
  - How do you change beliefs?

Propositional Knowledge (<math display="inline" xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mi>ω</mi><annotation encoding="application/x-tex">\omega</annotation></semantics></math>-knowledge):

- Knowledge of **why** something works.
  - It consists of scientific principles, natural laws, and systematic explanations.
    - In this case of yoghurt, it is the biological and chemical understanding of bacterial fermentation, enzyme activity, and protein coagulation.

Moykr: Prescriptive versus Propositional Knowledge

Prescriptive Knowledge (<math display="inline" xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mi>λ</mi><annotation encoding="application/x-tex">\lambda</annotation></semantics></math>-knowledge):

- **How**

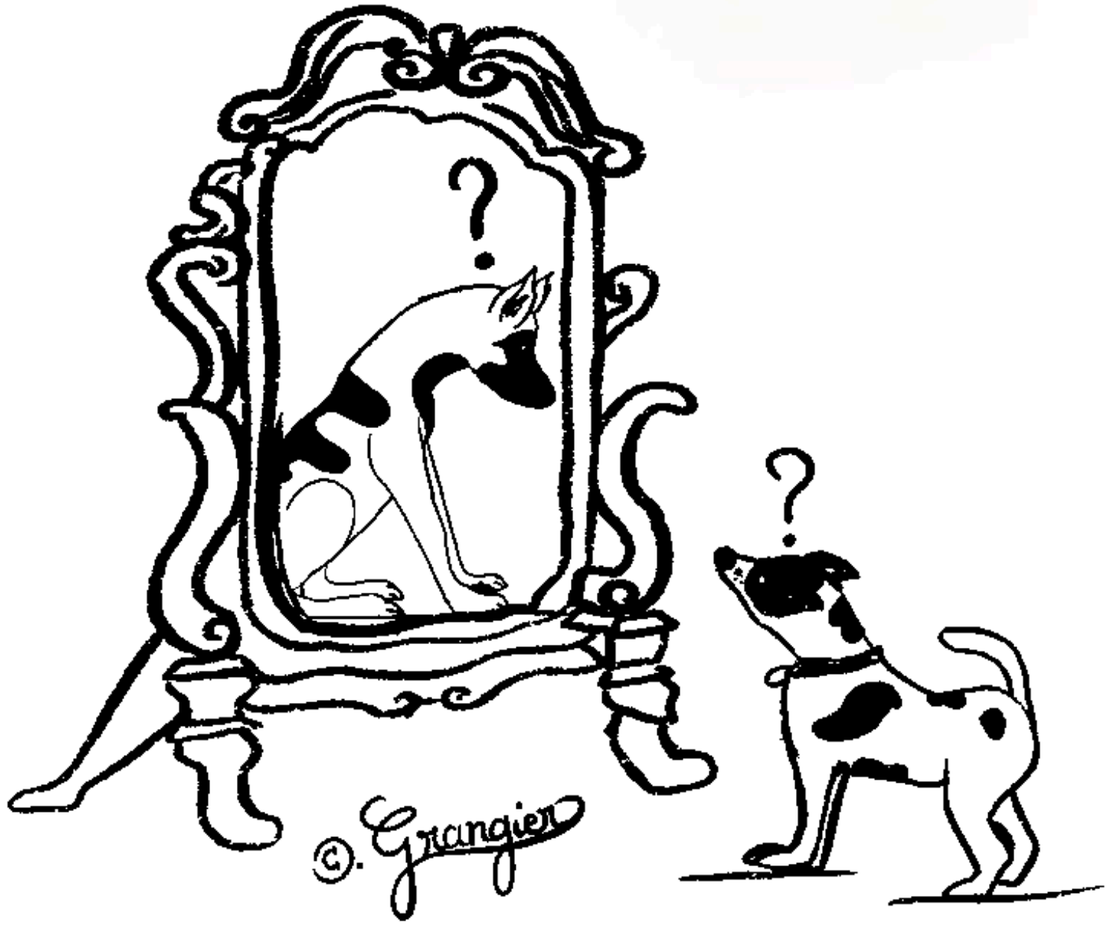


Propositional Knowledge (<math display="inline" xmlns="http://www.w3.org/1998/Math/MathML"><semantics><mi>ω</mi><annotation encoding="application/x-tex">\omega</annotation></semantics></math>-knowledge):

- **Why**



How would you define an academic literature?



Human + AI

Let’s forget about everything

we know about AI for a moment …

Three waves of technology

1960s–1980s

1. Computational Foundation Wave (Data Creation/Storage)
   - Silicon, Microprocessors, Relational Databases

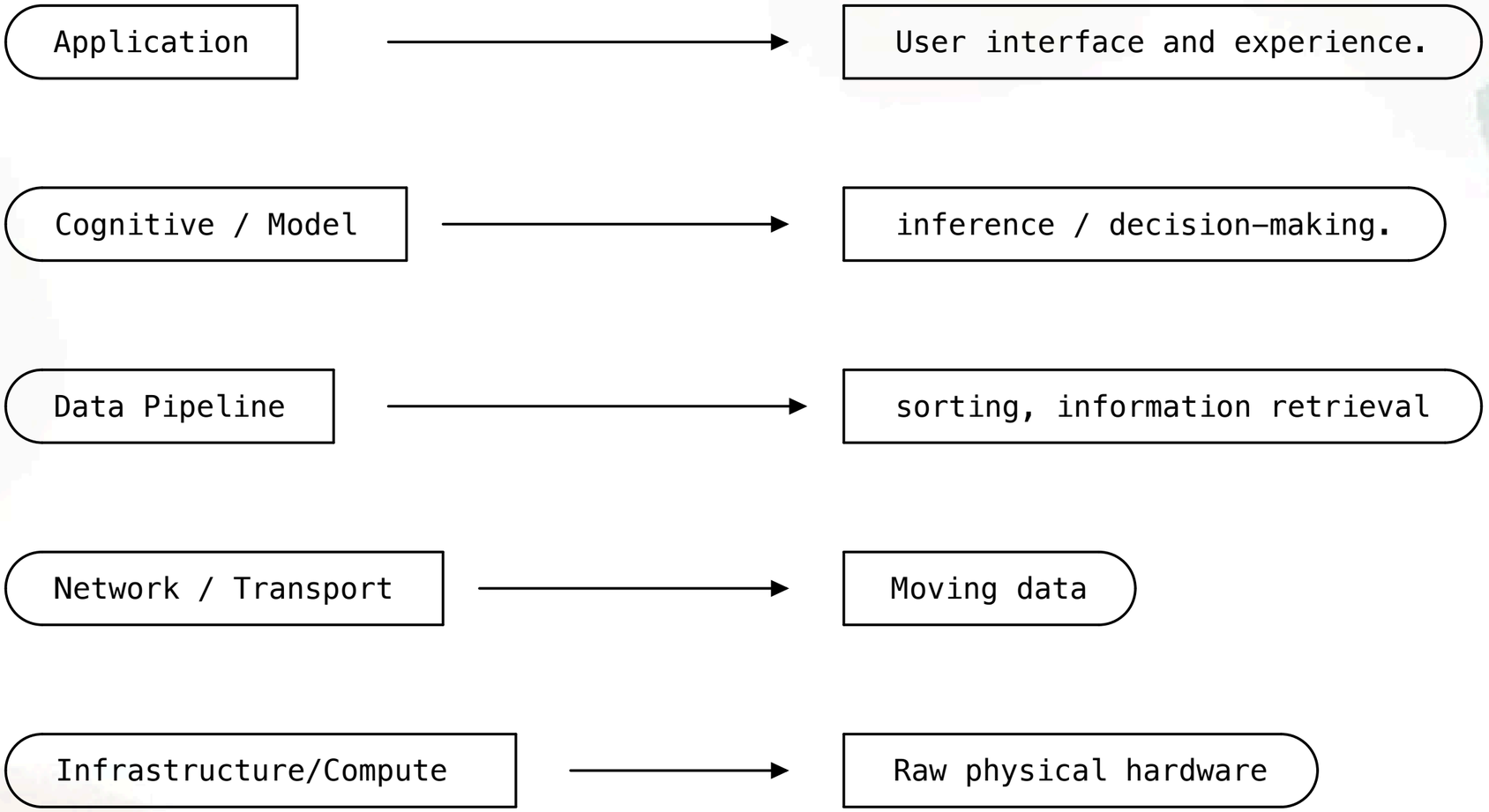
1990s–2000s

2. Connectivity & Transmission Wave (Data Flow)
   - Internet, TCP/IP, Cloud, Optical Fiber

2010s–Present

3. Cognitive / Intelligence Wave (Data Interpretation)
   - Neural networks, ML, LLMs

Layers of the AI Stack



Technology waves

Each wave created a layer that allows the next technological waves to operate

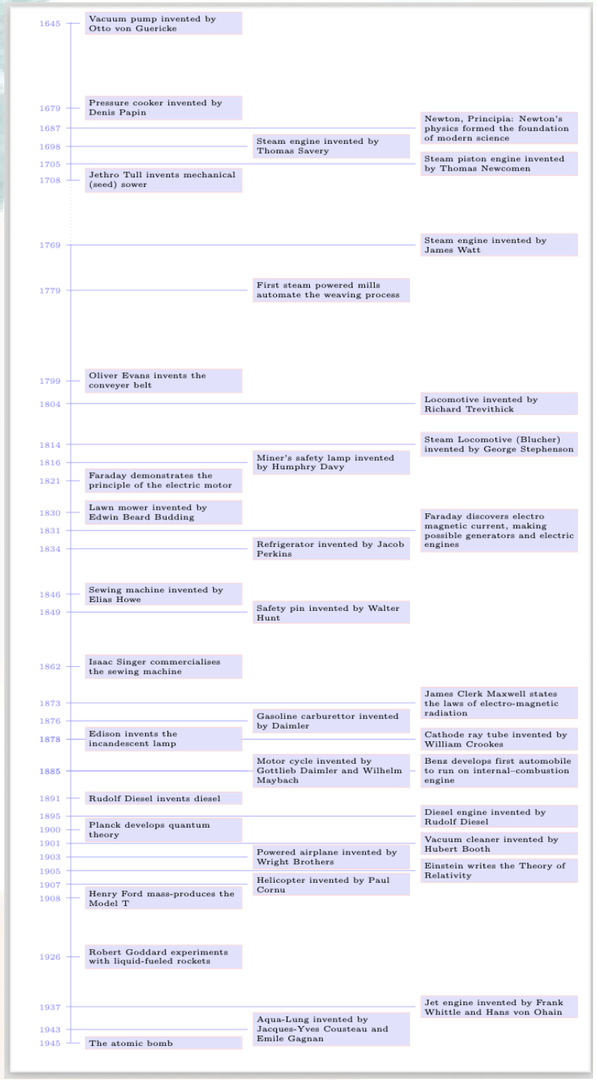
What is the relationship between the different layers

- dynamic complementarity

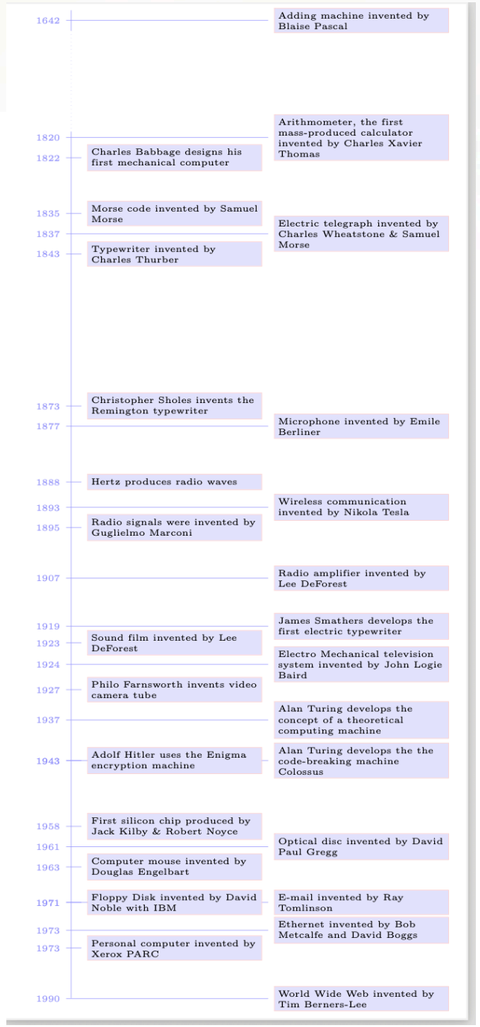
Dynamic complementarity

- complementarity
- dynamic

Evolution of Mechanical Inventions



Evolution of Computing Inventions



What AI can help with

1. Concept Explanation & Literature Mapping

- Tools can explain complex methods (e.g., ANOVA, mixed-effects models, p-values), suggest study resources, or help identify analytical frameworks used in published papers.

2. Specialised Scientific Discovery

- Platforms like Anthropic Claude Science help integrate fragmented databases, literature, and analytical tools in fields like biomedicine and formal sciences.

What AI can help with

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3. Data Cleaning & Preprocessing

AI can draft cleaning scripts, parse unstructured or mixed data, and identify structural issues—provided a human reviews the actual transformations.

4. Code Generation and Debugging

Instead of performing math directly, most modern AI tools write and execute code (such as Python, R, or Stata) to run data transformations and statistical tests.

Major Risks and Limitations

Hallucinations and Methodological Misalignment

LLMs predict text and code sequences based on patterns; they are not trained statisticians. They may apply the wrong statistical test to your data structure or misinterpret edge cases while sounding completely confident.

Causal and Contextual Blind Spots

AI struggles with deep causal logic, study design nuances, and qualitative context, often favoring overly complex or popular methods over appropriate ones (complexity and popularity bias).

Data Privacy and Security

Uploading proprietary, sensitive, or personal research datasets into public conversational AI instances risks data memorisation or leakage.

The Code Verification Bottleneck

As noted in analyses on Medium, using AI for data analysis means you are really reviewing the code it wrote; if you cannot read or audit the code yourself, you cannot safely trust the output.

Best Practices for Using AI Responsibly

Keep a Human-in-the-Loop

- Treat AI as a research assistant, not the principal investigator.
  - You are fully responsible for the methodology, ethics, and validity of the findings.

Inspect and Run Code Independently

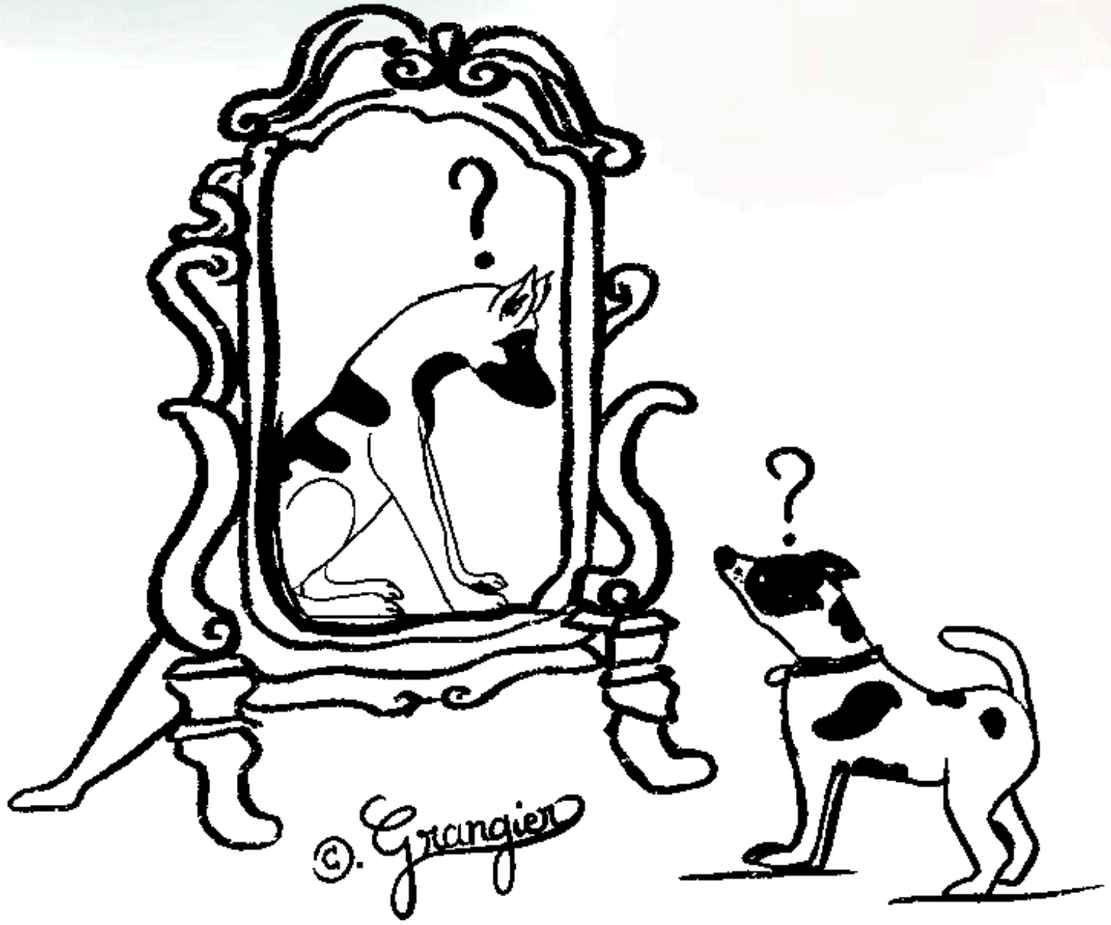
- Do not rely on black-box answers.
  - Have the AI generate clean scripts (Python/R), and execute and verify them outside the chat interface.

Use Prescriptive Prompting

- Avoid vague exploratory prompts that let the AI guess how to analyze your data.
  - Explicitly state the variables, hypotheses, and exact statistical models you want tested.

Protect Data Privacy

- Use local, open-source models on private infrastructure or institutional enterprise tools compliant with data protection policies when working with sensitive datasets.


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